Your Report Title Here

Muhammad Aitasaam Rana
Roll Number: 28100374
28100374@lums.edu.pk

Abstract

Write a short summary of the problem, what you did, and your main findings (150–250 words).

Code for all experiments is available at: `https://github.com/your-username/your-repo`.

1 Introduction

State the core question, motivation, and a brief overview of the tasks or experiments. Cite related work with [**?**] or **?**.

2 Experimental Setup / Methods

2.1 Task 1

Setup and models. We fine-tune `Qwen3-0.6B` on $40{,}000$ MetaMathQA examples (an LLM-expanded version of GSM8K and MATH problems) and evaluate seven checkpoints: (1) the untouched **Baseline**; (2) **Full FT Midpoint** (625 updates, $20{,}000$ examples); (3) **Full FT Full** ($1{,}250$ updates, 1 epoch); (4) **Full FT +** L_2**-SP**, full fine-tuning with L_2-SP distance regularization; (5) **LoRA-4** and (6) **LoRA-16**, standard LoRA with rank 4 and 16; and (7) **OPLoRA**, double-sided Orthogonal Projection LoRA (final checkpoint). Training sequences are capped at $2{,}048$ tokens without truncating target answers. To keep evaluation strictly zero-shot, any prompt shared by MetaMathQA and the GSM8K test set (after text normalization) is removed from training.

Evaluation. All models are evaluated zero-shot. **GSM8K** (math transfer) uses generation: the final numerical answer is parsed by regex, and unextractable outputs count as incorrect. Generating also lets us inspect response style and whether reasoning changes; the cap is $1{,}024$ new tokens, half the training cap. **HellaSwag** (commonsense) and **ARC-Easy** (science) use candidate likelihood, reporting raw (`acc`) and length-normalized (`acc_norm`) accuracy. **BoolQ** (binary reading) also uses candidate likelihood; since yes/no options have equal token length, only raw accuracy is reported.

Diagnostics. **Activation probes** compare each method's hidden-layer activations against the baseline on 64 fixed prompts (32 GSM8K test and 32 HellaSwag validation samples, seed 6304). A **qualitative non-math inspection** uses 24 fixed prompts from `first_plot_questions.yaml` (8 core questions in plain, JSON, and template formats), reporting instruction compliance, coherence, answer style, and inappropriate content.

Checkpointing. To stay under Kaggle's 20 GB disk limit, we kept only the most recent recovery checkpoint on disk, overwriting older ones as training progressed. For the final step-$1{,}250$ checkpoint we saved only the model weights and dropped the optimizer state, since it is needed only to resume training, not for benchmark evaluation or diagnostic probes.

Task 2: Name. Method description.

3 Results

3.1 Task 1: Name

Discussion of results, referring to Table **??** and Figure **??**.

3.2 Task 2: Name

Discussion of results.

4 Cross-task Discussion

4.1 Theme 1

Synthesize findings across tasks.

4.2 Theme 2

Further synthesis.

5 Conclusion

Summarize findings, state limitations (training budget, dataset size, seeds), and suggest future work.

References

A Task 1: Supplementary Material

Additional tables and figures.